THE UNIVERSITY OF MELBOURNE

Machine Learning (COMP30027) Tutorial Week 2

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Tutorial with May

Tuesday 10:00 AM
PAR- 104-L1-102

Wednesday 09:00 AM
PAR- 104-L1-102

Friday 12:00 AM
PAR-110-L3-3012

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Contact us

General inquiries: Ed forum on LMS

Personal/private concerns: Email the instructors

What‘s the workshops about?

First part – Tutorial, do the exercises sheet. Thinking & Discussion!

- Work through numerical examples with me
- Understand how and why the algorithms work

Similar to the type of questions you’ll see on the **final exam**

Second part - Coding exercises

- Time to practice **on your own**

What‘s the tutorial about?

COMP3002... > Modules > Syllabus > Week 1. Intr...

Immersive Reader

2024 Semester 1

- Home
- Subject Overview
- Modules
- Assignments
- Ed Discussion
- Quizzes
- Lecture Capture
- Zoom
- Grades
- Gradescope

Week 1. Introduction to machine learning

In this week's lectures, we'll introduce the subject and basic concepts of machine learning. What does it mean for a computer to learn?

Note: The related tutorial and practical will occur in the week following this lecture, and the solutions will be posted at the end of that week (so about two weeks after the lecture).

Lectures

1. Introduction [slides]

2. Basics of machine learning [handout]

Workshops

Conceptual Questions: Machine learning basics [handout] - [solution]

Coding Exercise: Reading and Processing Data [handout + data] - [solution]

WHAT IS MACHINE LEARNING?

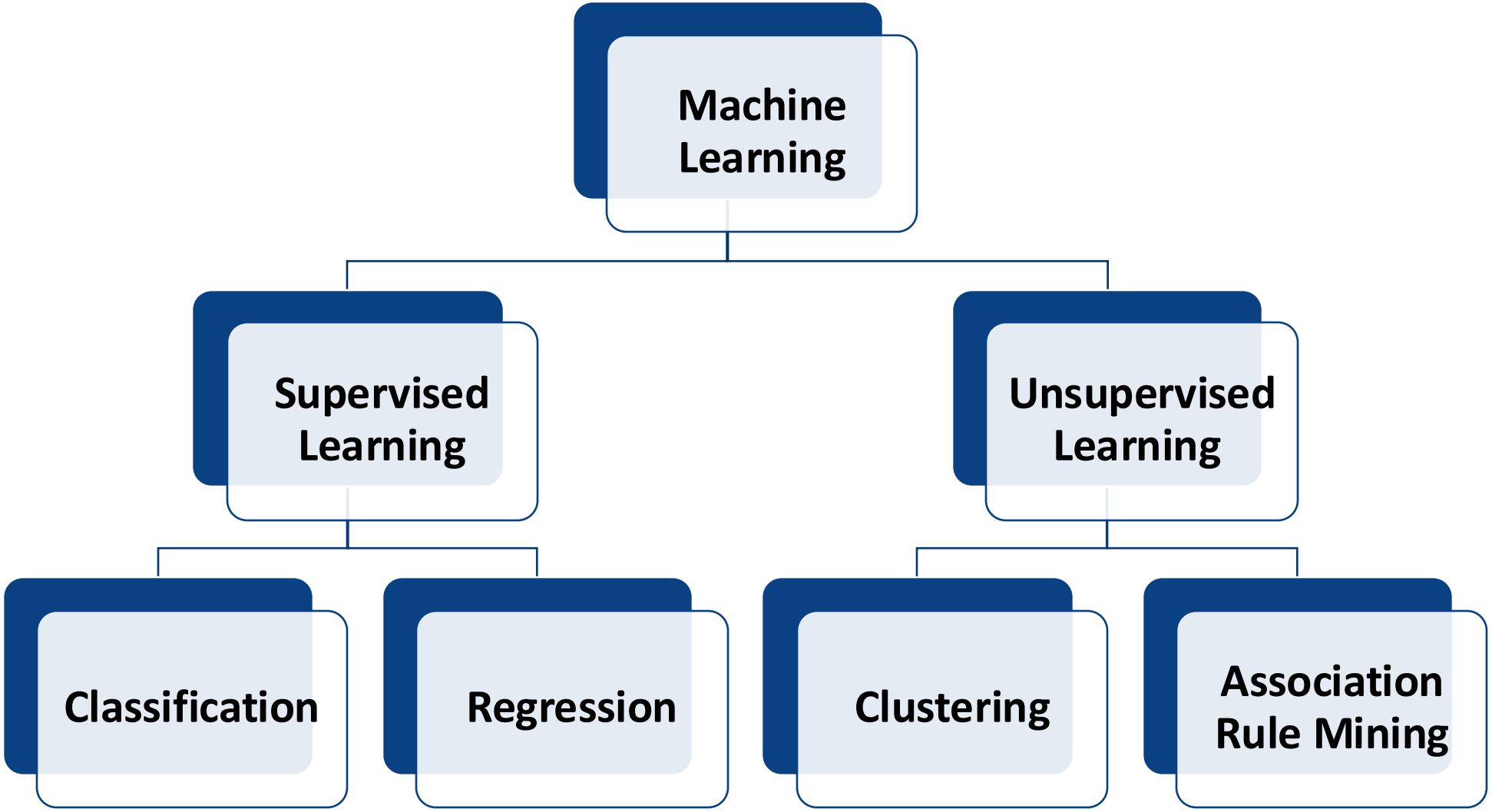
What is Machine Learning? (Video)

Video : What is Machine Learning?

Source: YouTube OxfordSparks

Link: https://youtu.be/f_uwKZIAeM0

Today's mission



Exercises

Considering the following problems:

(i) Skin cancer screening test.

(ii) Building a system that guesses what the weather (temperature, precipitation, etc.) will be like tomorrow.

(iii) Predicting products that a customer would be interested in buying, based on other purchases that the customer has previously made.

Exercises

1. Identify the "concept" we might attempt to "learn" for each problem (Task Identification).

Considering the following problems:

(i) Skin cancer screening test.

(ii) Building a system that guesses what the weather (temperature, precipitation, etc.) will be like tomorrow.

(iii) Predicting products that a customer would be interested in buying, based on other purchases that the customer has previously made.

Exercises 1. Identify the "concept" we might attempt to "learn" for each problem.

(i) Skin cancer screening test.

Is this A or B?

(Binary) Classification Algorithms

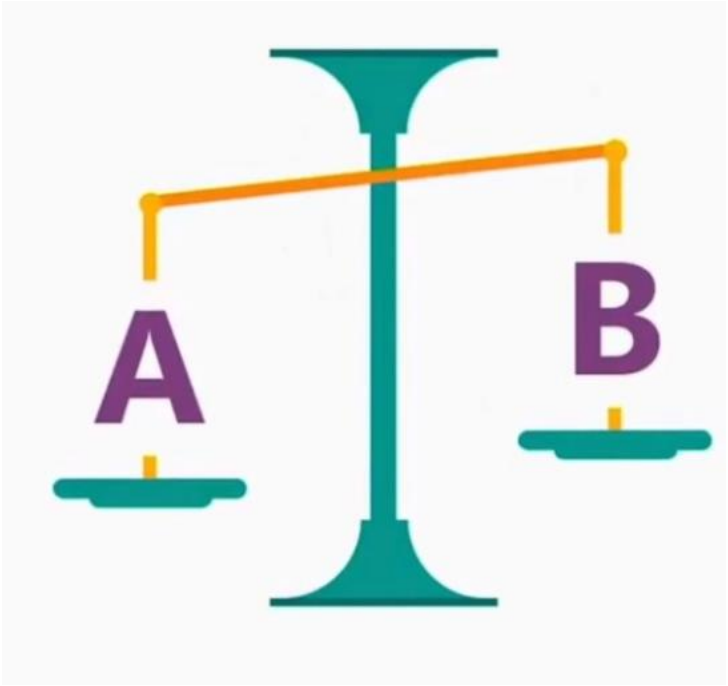


*Image source: Microsoft

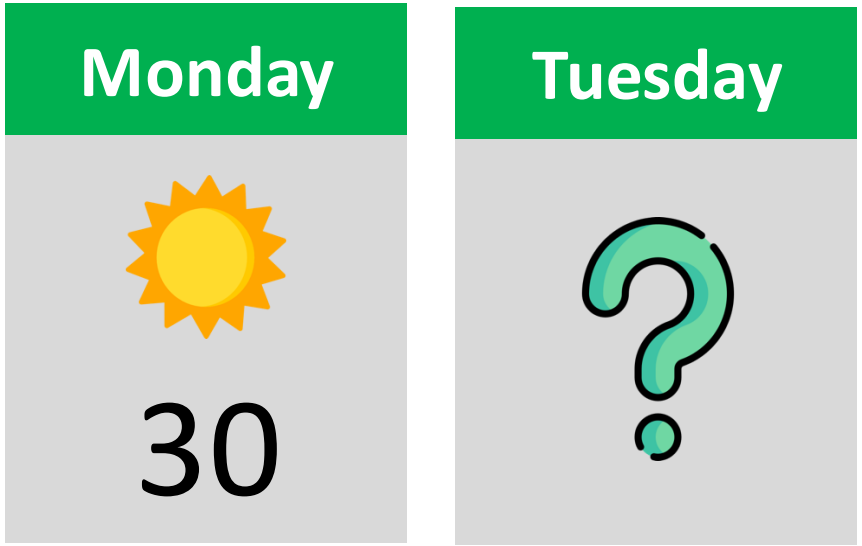
Exercises 1. Identify the "concept" we might attempt to "learn" for each problem.

(ii) Building a system that guesses what the weather (temperature, precipitation, etc.) will be like tomorrow.

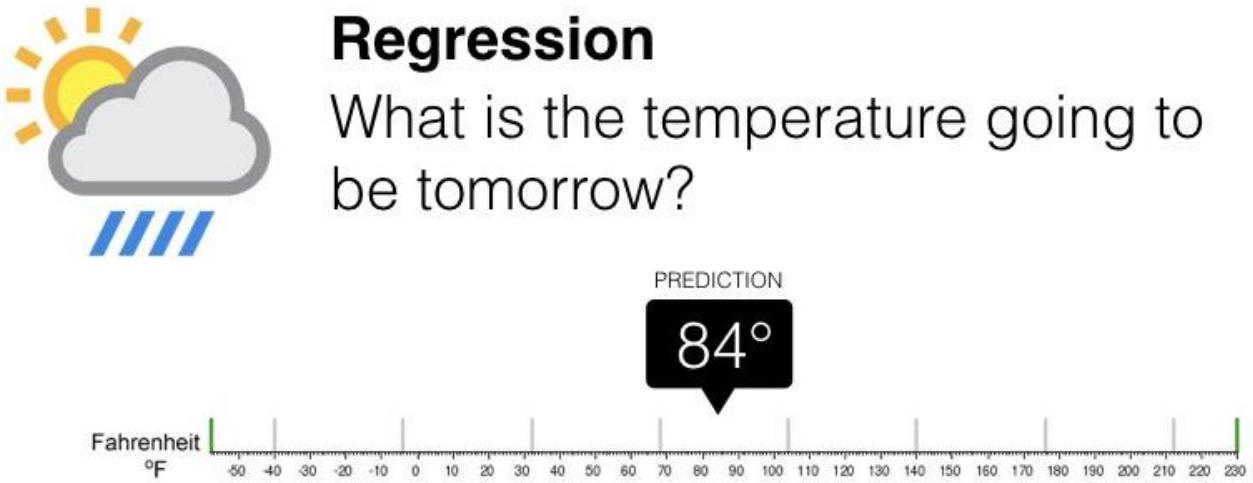
How much? How many?

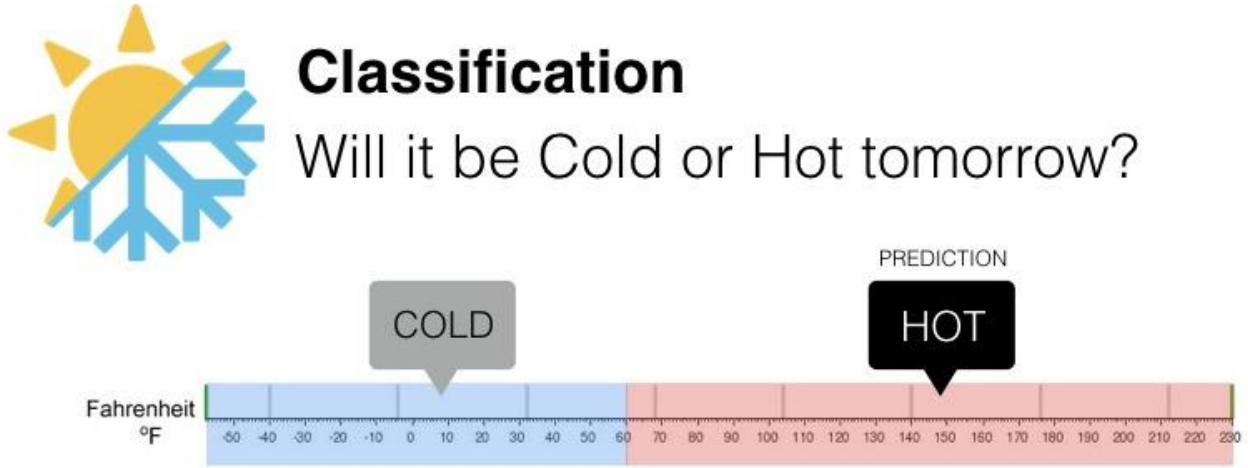
Make numerical prediction

Regression Algorithms



Classification vs. Regression

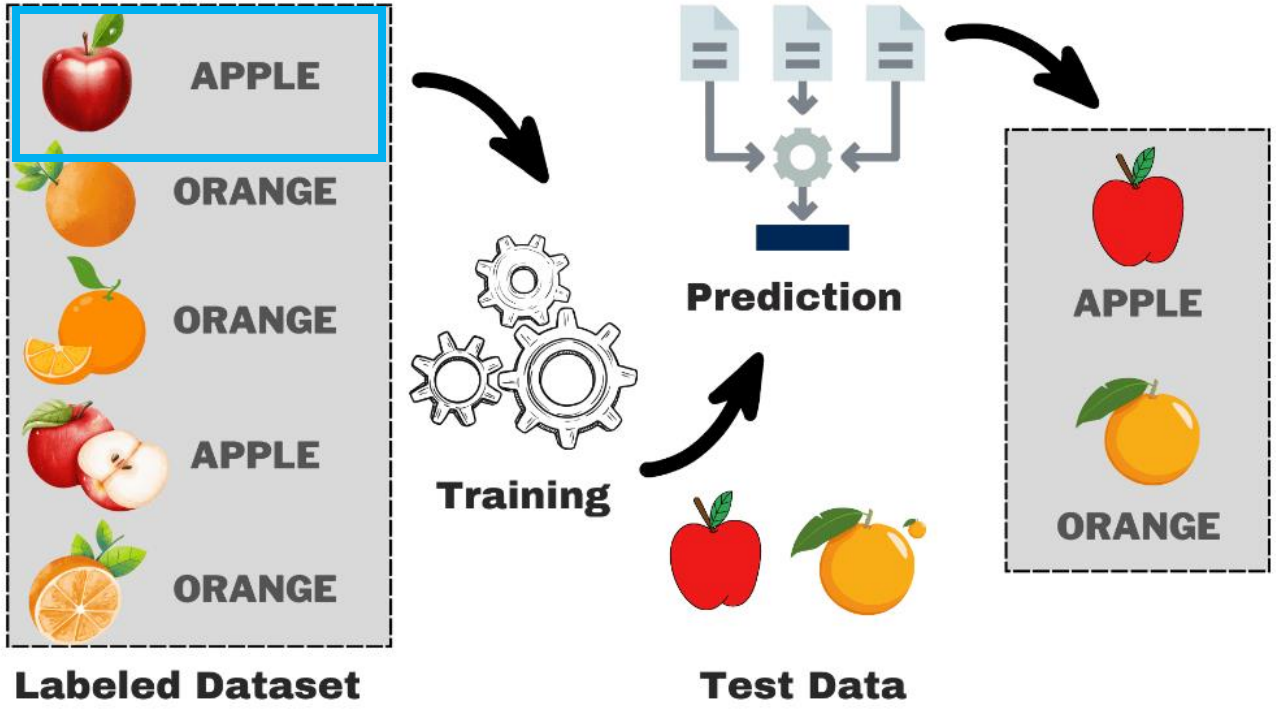




Supervised Learning

- Supervised learning is an algorithm that predicts the outcome of a new input based on **known pairs (data, label)**.
- Have a set of input variables X={x1,x2,…,xN} and a corresponding set of labels Y={y1,y2,…,yN}, with xi, yi are vectors.

(data, label) = (x1, y1)



Exercises 1. Identify the "concept" we might attempt to "learn" for each problem

(iii) Predicting products that a customer would be interested in buying, based on other purchases that customer has previously made

Is this A or B?

(interested or not interested)

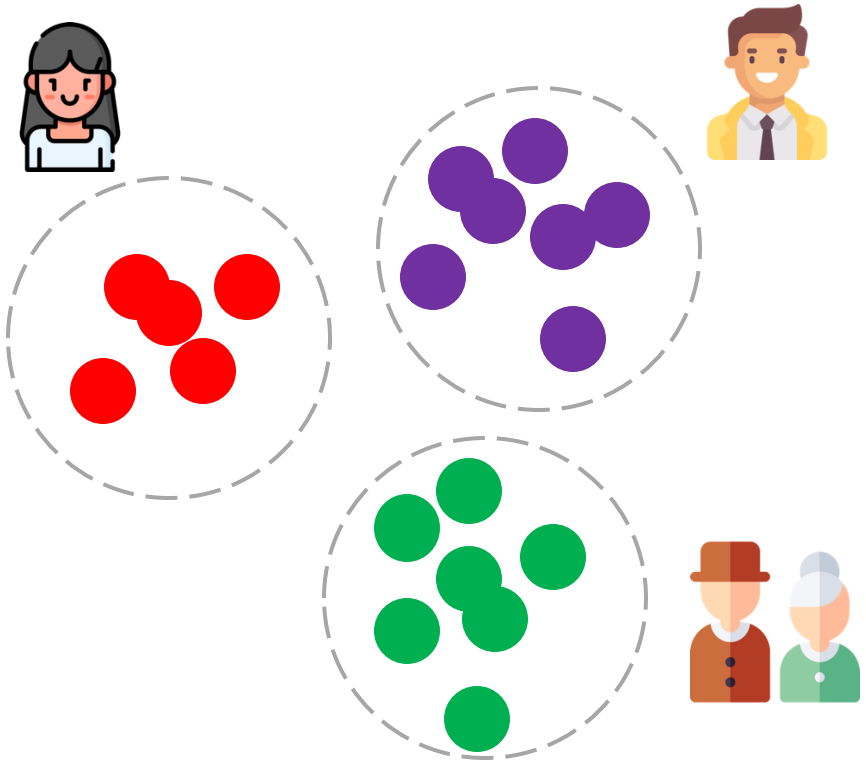
Classification Algorithms

Exercises 1. Identify the "concept" we might attempt to "learn" for each problem

(iii) Predicting products that a customer would be interested in buying, based on other purchases that the customer has previously made

How is this Organized?

Clustering Algorithms



Exercises 1. Identify the "concept" we might attempt to "learn" for each problem

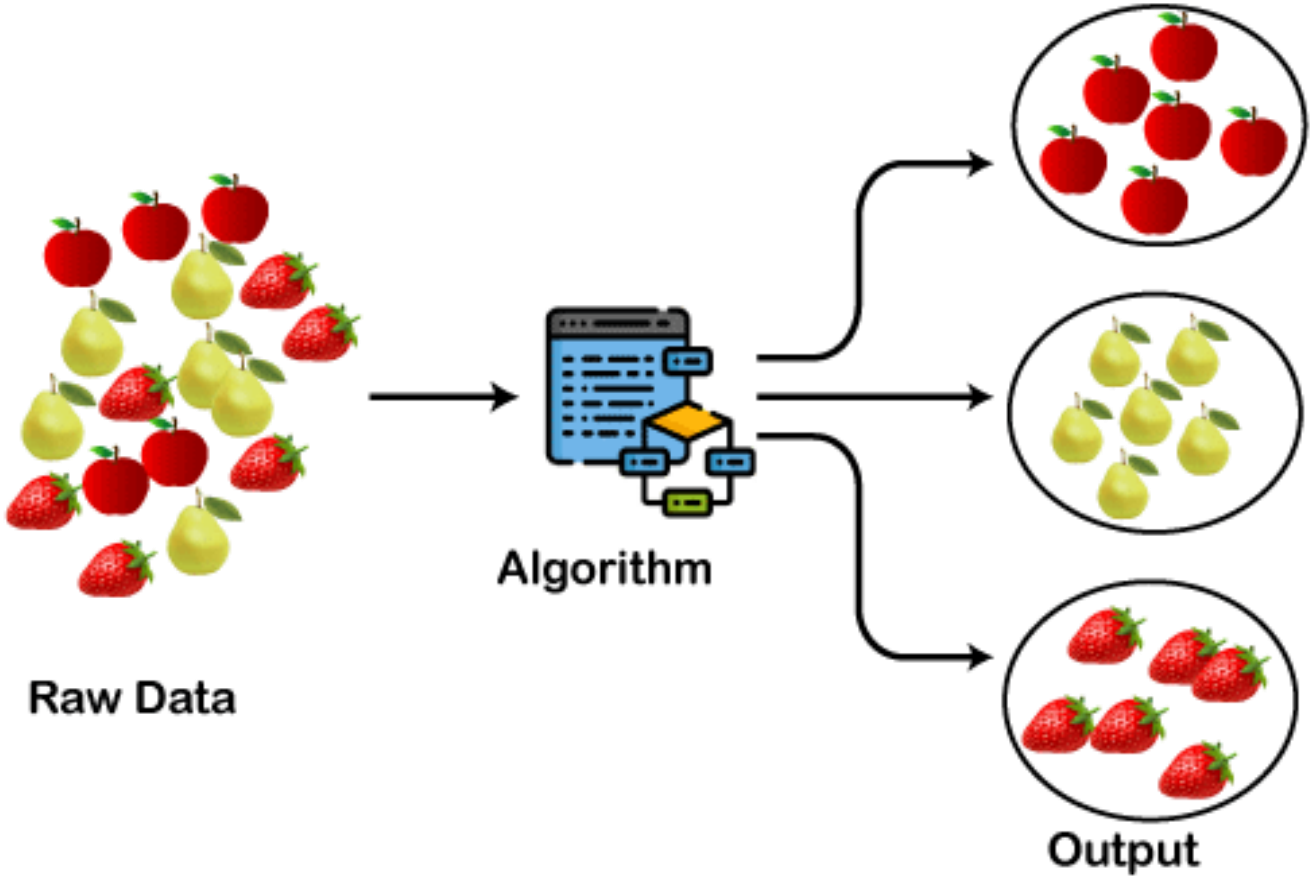
(iii) Predicting products that a customer would be interested in buying, based on other purchases that customer has previously made

How or Why certain items are connected?

Association Rule Mining

Unsupervised Learning

- **Do not know** the outcome or **label** but only the input data.
- Only have input to X without knowing the corresponding Y label.

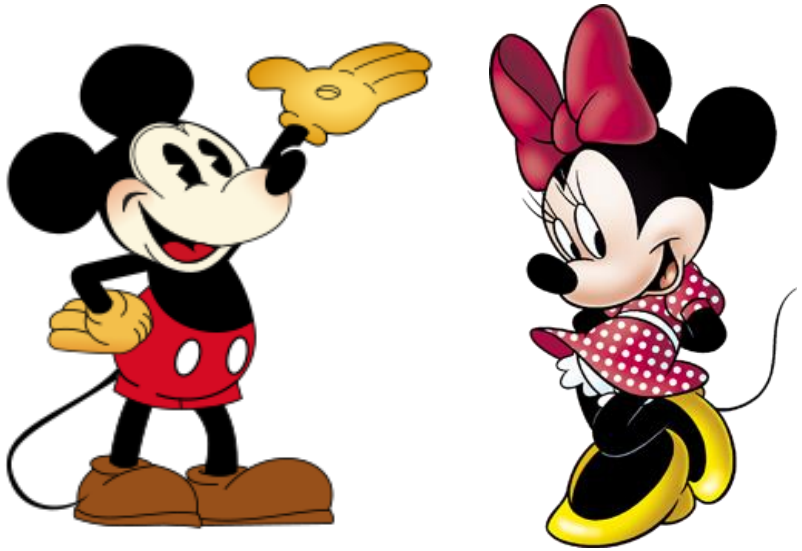


Instance and Attribute

Class: **Mouse**

Instance: **Minnie** **Micky**

Attribute: **name, age, gender**



Exercises 2. For each problem-task, identify what the instances and attributes might consist of (choosing the data representative)

(i) Skin cancer screening test.

- Instance: patient.
- Attribute: blood test, images from the skin, reports, observed syndromes,…

(ii) Building a system that guesses what the weather (temperature, precipitation, etc.) will be like tomorrow.

- Instances: a day.
- Attribute: temperature, precipitation, humidity, wind speed,…

(iii) Predicting products that a customer would be interested in buying, based on other purchases that the customer has previously made

- Instances: customer-product pairing/ customer.
- Attribute: name, age, address, gender, shopping log, credit card information,…

Exercises 3. For each problem-task, conjecture whether a typical strategy is likely to use supervised or unsupervised Machine Learning (picking a suitable model)

(i) Skin cancer screening test.

Classification

(ii) Building a system that guesses what the weather (temperature, precipitation, etc.) will be like tomorrow.

Classification

Regression

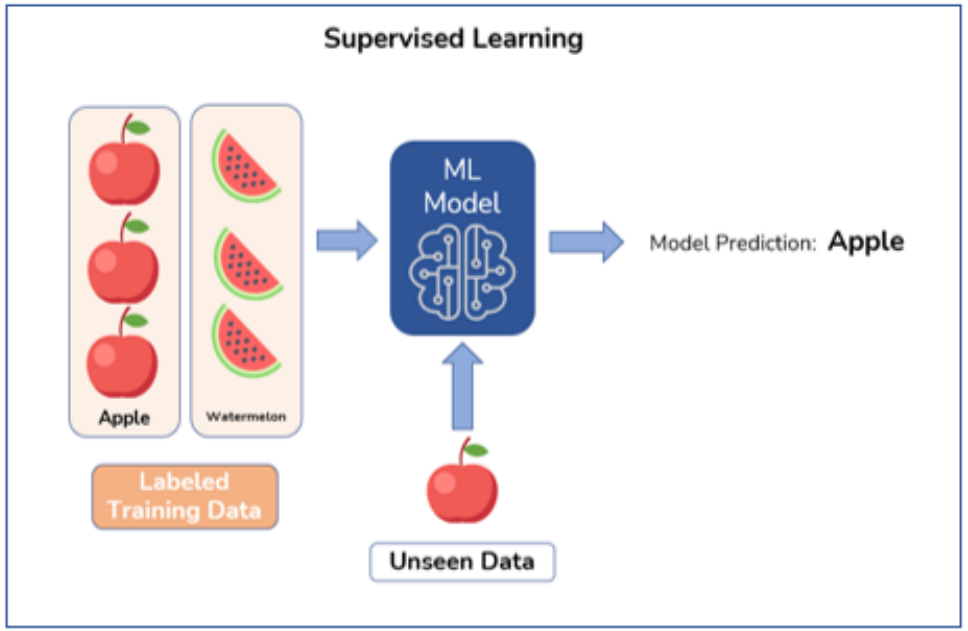
(iii) Predicting products that a customer would be interested in buying, based on other purchases that the customer has previously made

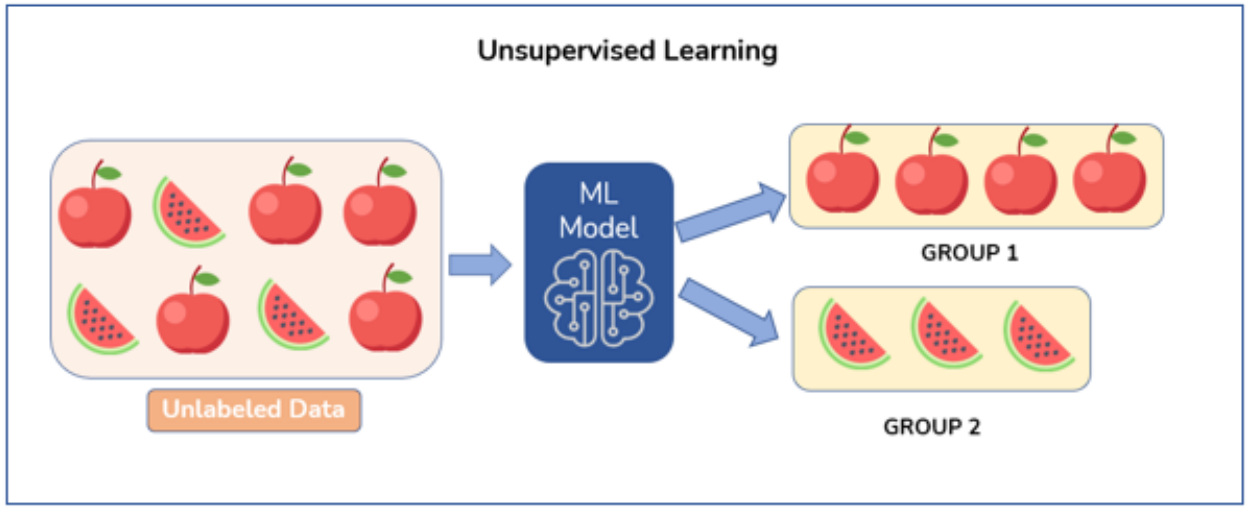
Classification

Clustering

Association Rule Mining

Supervised or Unsupervised Learning





Supervised: start from exemplars (instances) - labeled with classes - in a set of training data and use these to classify unknown instances in a set of test data.

Unsupervised: not based on a set of labeled training data.

Often broken down into *'weakly unsupervised methods'* (where the class set is known, but the system does not have access to labeled training data), and *'strongly unsupervised methods'* (where even the class set is unknown, and we don't even know how many classes we have).

What is Machine Learning? (Video)

Video: Supervised vs. Unsupervised Machine Learning:

What's the Difference?

Source: YouTube Eye on Tech

Link: https://www.youtube.com/watch?v=rHeaoaiBM6Y

Discussion

4. For each problem-task, consider how easy or difficult it would be to make a model that generalizes to new cases.

5. What kinds of assumptions might a machine learning model make when tackling these problems?

4. For each problem-task, consider how easy or difficult it would be to make a model that generalizes to new cases.

(i). Skin cancer screening test.

Generalization 🤔 Real world training data often have biases.

For example, we know skin cancer risk increases with age, so these variables will probably be correlated in your training set.

👍 the model should correctly learn that age predicts skin cancer

👎 the model becomes too dependent on that predictor

4. For each problem-task, consider how easy or difficult it would be to make a model that generalizes to new cases.

(ii). Building a system that guesses what the weather (temperature, precipitation, etc.) will be like tomorrow.

Might work better in some cities than others.

Probably generalize better if it included geographic information.

4. For each problem-task, consider how easy or difficult it would be to make a model that generalizes to new cases.

(iii). Predicting products that a customer would be interested in buying, based on other purchases that customer has previously made.

A customer model trained in one country might not generalise to other countries.

If it mostly learns everyday shopping patterns, it probably won't give good predictions for outlier situations like holiday purchasing.

5. What kinds of assumptions might a machine learning model make when tackling these problems?

Every model makes assumptions about the world and how the concepts we want to learn relate to the attributes of the data.

- The concept is actually related to the attributes.
- Each model makes assumptions about the ways the attributes can relate to the concepts.